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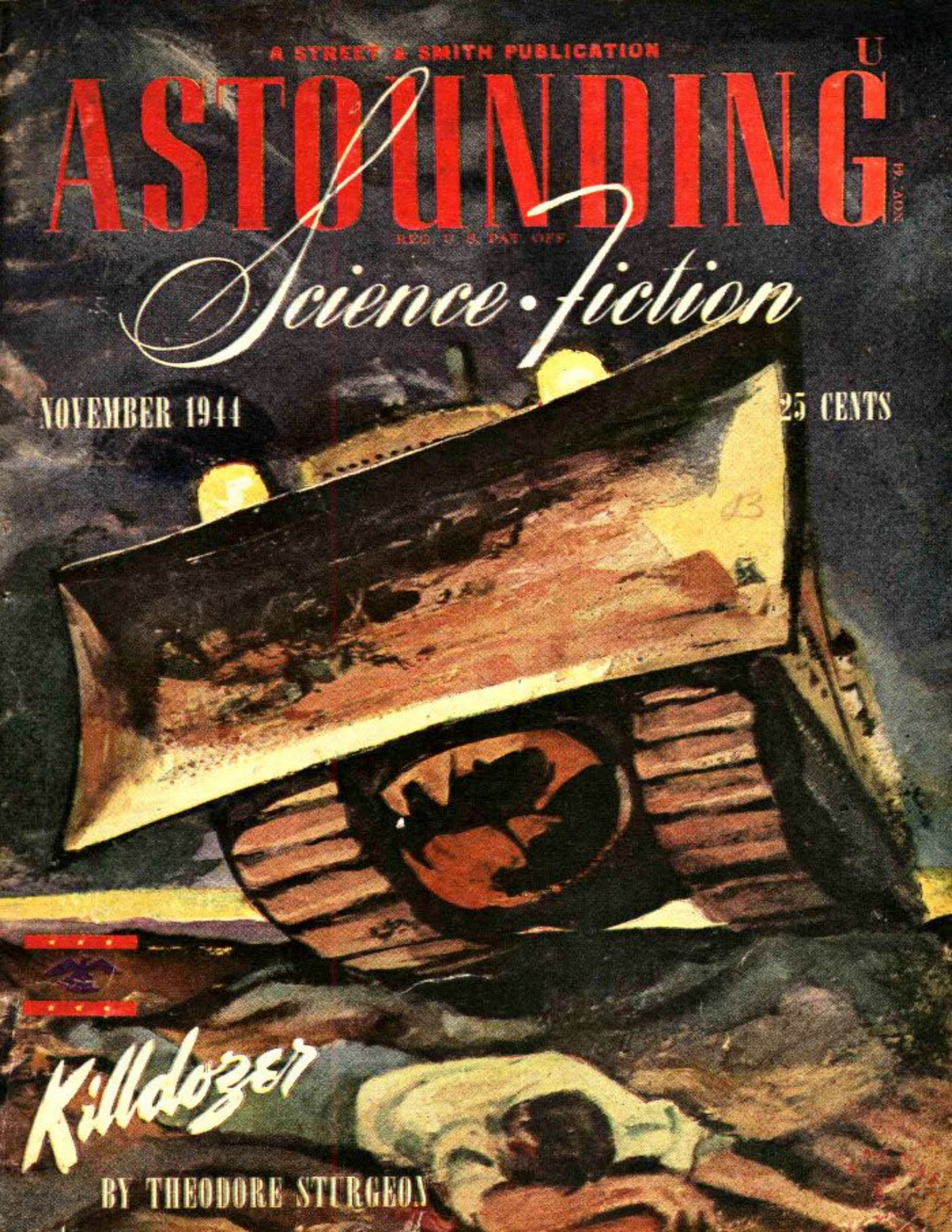
Science Fiction

NOVEMBER 1944

25 CENTS

Killdozer

BY THEODORE STURGEON



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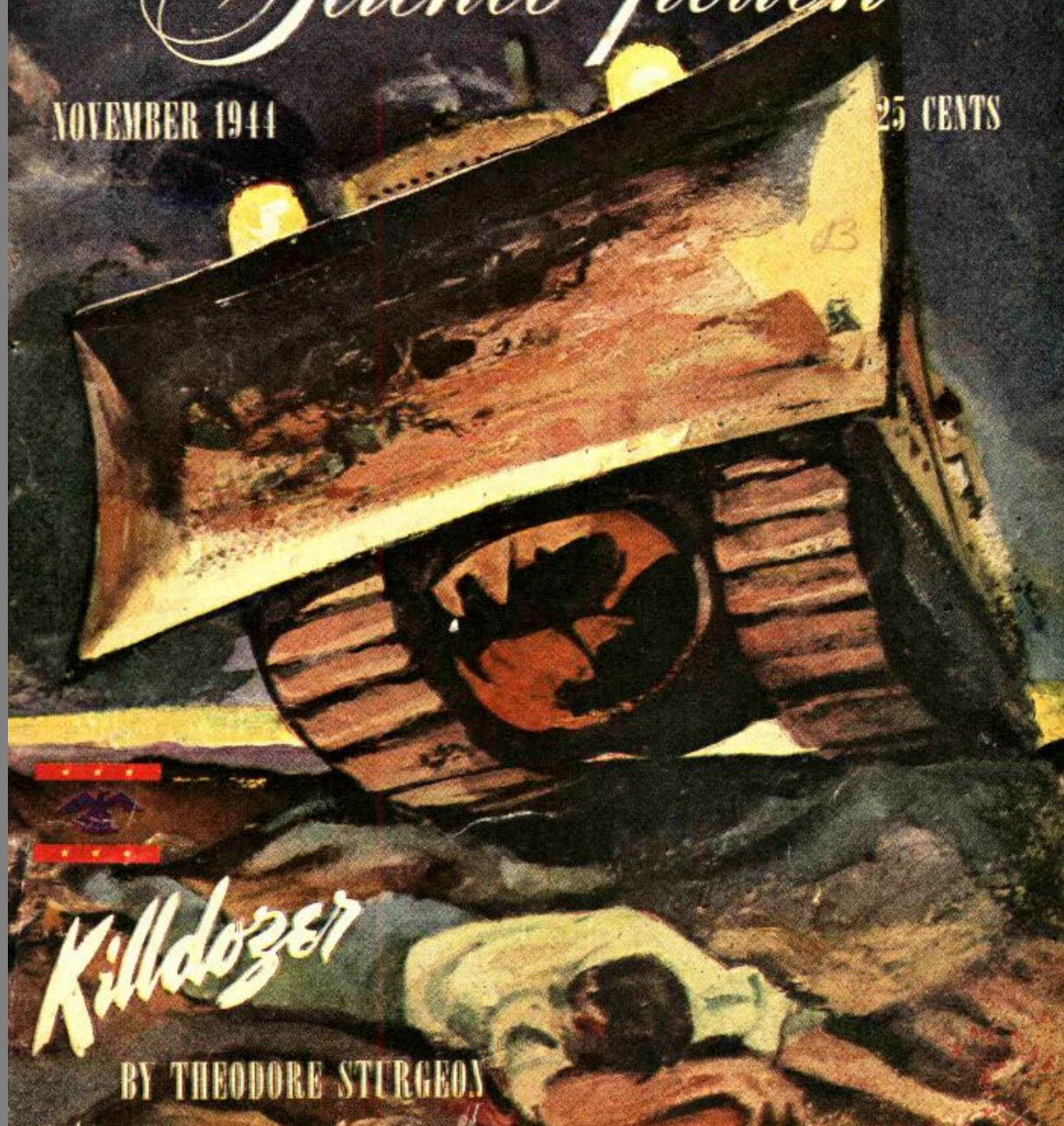
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*** START OF THE PROJECT GUTENBERG EBOOK
REDEVELOPMENT ***

Redevelopment

By WESLEY LONG

Illustrated by Williams

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John McBride hung the phone on the hook and wiped his face. This face-wiping was not the usual gesture of a man whose face is dirty, or covered with perspiration. It was the dazed sort of gesture made by a man who has just been subjected to a surprise, and since the wiping tended to remove the awed look, replacing it with a slightly dazed smile, the surprise must not have been too unpleasant.

He shook his head, as though to clear it, and then made his way through Station 1 of the Plutonian Lens to the landing platform. Just inside the gigantic lock, a medium-sized space-ship stood, and sitting on the edge of the space lock, swinging her feet, was Sandra Drake.

"Hello," she said brightly.

"Hi," said John. This was entirely new. Sandra Drake was not usually given to greeting men as anything but absolute imbeciles. "What brings you out here? And how did you make it?"

"Oh," said Sandra lightly, "I remembered the charge on Station 1 and brought along a charge-compensator. We hardly sparked when we lit."

One of the attendants said, in a low aside: "About three hundred amperes! She'd call a major explosion a snap of the fingers! You could hide an egg in the crater she made."

But Sandra was still talking. "John," she said in a voice that would have caused Shylock to give her his last gold piece, "I want help."

"You need help? What can we do for you?"

"It's pretty big," warned Sandra. Her low contralto dared him to ask what it was—and also dared him to deny it to her.

"Look, Drake, you did us a favor not too long ago. I think we owe you one."

Sandra smiled uncertainly. "I was afraid that that little stunt was only repaying you for the first meeting we had."

"Shucks," said McBride. "Anyone can make a mistake. Forget it."

"But being pilot for you on the *Haywire Queen* did me a lot of good, too, you know. I got my license back for that one. We both gained."

"I know. I'm glad we did. But what can you possibly want that is so big that you're afraid to ask?"

"Well, and maybe it isn't too big, either. Steve is a friend of both of us, isn't he? I'd do anything for Steve—and wouldn't you?"

"Yes. If any favors are owing, I think it is both of us to him."

"That's what I'm getting at. I need help—for Steve."

"You sure go a long way around to get it," grinned McBride. "Why didn't you tell me that first instead of warning me about a favor?"

"It's pretty big. But look, John, Steve took the *Haywire Queen* on a run to Sirius more than six weeks ago. He took along enough stuff to stay a week;

he said he'd be back after one hundred and seventy hours of stay at, on, or near Sirius. This was just a trial hop to try the new drive you cooked up and a longer, better equipped expedition would be made later."

"He did say something about it the last I saw him. He said he wasn't particularly interested in exploring a new system. He'd leave that for the explorers. He was interested in the drive and so on, and after he'd paved the way for getting to the stars and had proven his drive, he'd turn it over to those interested in colonization. But six weeks ago, you say? Gosh, that's a long overstay, isn't it?"

"It is. I happen to know he didn't take more supplies than he needed. So I'm worried about him."

"And where do I come in? You want me to go and help you look for him?"

Sandra smiled wanly. "Hardly. I'm sure Enid would enjoy that, too. No, John, what I want is for you to hook up the stuff I've got in the *Lady Luck* to make me one of those drives you invented so that I can go myself."

"You're taking a chance, you know."

"That's where the favor part comes in. I want to go and look for Steve Hammond. I need your drive. And if you don't help me, I'll go out in space and tinker with the junk until I get it. I was there when you cooked it up, remember, and I have a good memory for details."

"But it's dangerous."

"Is it? 'Might be dangerous' is what you mean. And I've been taking harebrained chances for a long time, now. Do I or don't I?"

McBride thought for a long time. "You get it," he said at last. "On one condition. That you return in not less than one month. If you do not, I'm going to take it upon myself to follow. So no matter what you find, get back. Is that a promise?"

"It is."

"O.K., Sandra." McBride went to the wall of the big lock and spoke over the communicator. "Tommy! Get Al and Westy and tell 'em to bring their tools to the landing lock. We're going to juggle a few generators around."

To Sandra, he said: "I hope you've got plenty of what it takes."

"I have," she said, sensing his meaning. "Matter of fact, I've got the latest thing in alphas—two of 'em. And all the E-grav generators we'll need are all tacked into what I think are the right places to make this crate into a super-speed job. There are spares for all three fields, and a couple of spare cupralum bars, too. Even part of the wiring is done. I got just so far and then realized that I don't know too much about gravitics. That's when I decided to come here for help."

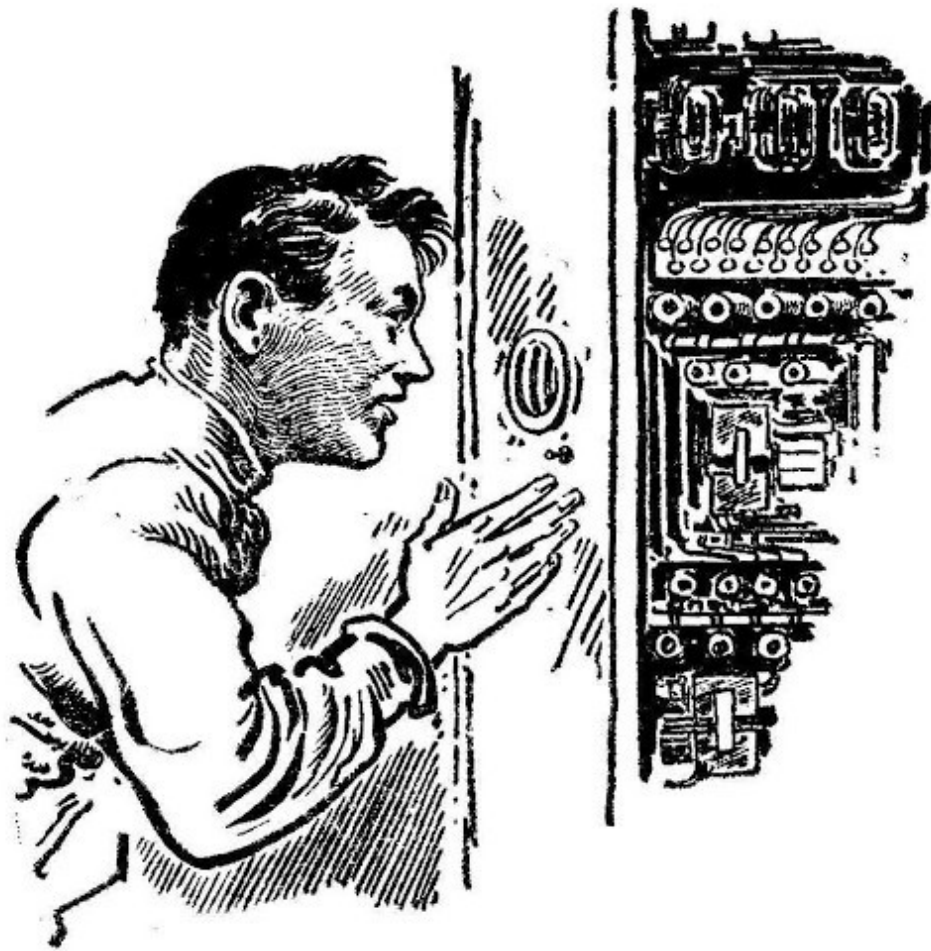
"Good thing," said McBride. "You might have killed yourself."

Sandra didn't answer, and at that moment, McBride's men came with their tools. Wordlessly, they nodded to Sandra and then followed McBride into the *Lady Luck*.

McBride wasted no time. "Al," he said, "you fit the mag-G for vertical bi-lobar field to cover the nose of the crate with the top lobe, and Westy, you see that the mech-G generator in the nose induces the proper vectors in the cupralum bar. I'll get Hank and Jim to touch up the wiring and safety devices. We'll have this crate back in space within the hour!"

"Working a little fast, aren't you?" asked Sandra.

"No. I don't think so. You've got most of the main stuff in place. It's merely a matter of running the alphas correctly—remember, Sandra, alphas are not electrons and even low-alpha lines require smooth, round bends, otherwise they squirt off in a crackling alphonic discharge that will eat the side out of a steel tank. You've done most of the heavy work. It just requires touching up here and there: getting the proper field-intensity out of the gravitic generators and adjusting the output of the alphas. Then there is some tricky relay work with the safety circuits: it wouldn't improve your beauty to suddenly find yourself sitting in the pilot's chair at seven thousand gravities."



Sandra shuddered.

"Oh, and look, since you've got the compensator. You'll find a static-charge meter handy, perhaps. If there are planets around Sirius, who knows what their intrinsic charge is. We'll loan you one so that you can make planet without making a corona at the same time. Rarefied air makes pretty lights when it comes under a few trillion volts—and being a cathode is no worse than being an anode when your voltage is running up into a bushel of zeroes—either is equally disconcerting. How do you intend to spot any planets?"

"I've got a pair of hemisphere lenses. I'll sail through the Sirian sky at about forty thousand miles per second and expose for ten minutes. The stars will

still appear as spots, but anything close enough to be planet-wise will make streaks unless it is dead ahead.

"In which case you'll see it personally," grinned McBride. "That's the best stunt I've heard of yet to find planets."

"It isn't new. They used it to see if there were any planets outside of Pluto several years ago, though they exposed for several hours while running at ten or fifteen thousand. Steve has a pair of hemis with him, too."

Al came trudging in with a roll of alphon cable over his shoulder and dropped it on the floor. "She's in—my end, anyway."

"Running already?"

"On test power. Drake had the bi-lobar field almost on the ball. Westy found about the same thing. I think another couple of days and Drake wouldn't have needed help."

"I couldn't make it work," complained Sandra.

"Well, you missed a few minor points," said Al. "Never, never run alphon lines anywhere near a relay rack. It induces crosscurrents in the windings and either makes 'em more sensitive or almost dead, depending on the polarity. It won't hurt AC relays, but they aren't used too much on a spaceship, so it's best to play safe."

"I'll remember that, too," Sandra promised him.

"O.K."

And so an hour passed, and another one added to it before the *Lady Luck* was fitted for super drive. It was finished, then, and Sandra Drake was more than voluble in her thanks.

"Never mind the thanks," said McBride, "or we'll be into that original wrangle as to who owes who what kind of a favor. Where we sit out here in the lens, favors are not weighted and set down as an asset. Forget it. G'wan out there and get Steve Hammond—and do not forget for one minute I'm

coming after you if you're gone more than thirty days. Seven hundred and twenty hours! Get me?"

"Sure thing," said Drake. "And, John, you're pretty swell."

"Nuts!"

"All right, 'Nuts!' But some day I'm going to settle down and be a good girl, and then you can believe me."

"That, I'll believe when I see it. Go on, Sandra, go out and get Steve."

"I'll get Steve," promised Sandra. "Oh, but definitely."

"Well, good luck."

"Thanks."

The space lock closed, and the men retreated inside of the Station's air lock. The gigantic doors swung open, letting a huge puff of air out into space. Then the *Lady Luck* lifted gracefully for all of her tons of mass, and wafted out through the opened door. It was a dead-center passage, one that could be made only with a master pilot running the board personally.

Then she was gone. Halfway around the lens she would have to go before Sirius came into a safe line of flight. Sandra was taking no more chances on contacting the surface of that mighty space-warp that focused Sol on Pluto.

McBride wondered: *Has Sandra learned her lesson?*

One week passed. One week, filled to the very brim with all of those routine things that make life full of wonder—as to whether there isn't something better in the hereafter. The sheer millions of miles of gravitic-induced space-warp refracted Sol's light endlessly and perfectly to make for Pluto a synthetic sun that sported a dozen darting points. On Pluto, men lived and worked and pursued happiness, and the valuable ore came up from the ground in the Styx Valley and created the need for Pluto and the lens. Over Mephisto, the smelters cast their glow against the sky, which the inhabitants of Hell always called "The Eternal Fire." Across the River Styx from Hell, Sharon lay like a city of marble by day and a string of pearls by night.

Nor was Hell, as seen from Sharon, any less beautiful. The twin cities of Pluto, rivals in everything, fought as usual. And the bone of contention for that particular week was a simple, age-old epithet. It is a sorry fact that with the entire solar system running as it always did, Sharon and Hell found it possible to make the headlines of all the cities of the system by their arguments.

Sharon lost. Hell succeeded in bringing to mind the fact that Hell, Pluto, was a fine place to be, and the poor citizens of Sharon were forced into second consideration. But then, Sharon had not been a running business for centuries.

Go to Sharon! had no familiar ring.

But the Road to Hell was a broad highway.

McBride looked up as the door to his office opened, and his jaw fell away down to here. He blinked. He looked again, and then jumped to his feet. "She found you!" he said.

"Who found who?" asked Steve Hammond. "Has that dame—?"

"Drake? Yep. She came here and we fixed that drive for her. She's changed, Steve. Even I can see it."

"So she was here?"

"You bet. Sandra has changed."

"Has she?"

"Why, Steve, she was actually worried about you. Near frantic."

"Was she?"

"She may have concealed it from you. After all, she's been a pretty hard-boiled girl and the change is a little abrupt. She's probably concealing her real feelings."

"Would she?"

"Probably. After all she's said about men in general, she's probably fighting an internal battle. But she let it go right here."

"Did she?"

"Did she! Why, she tried to hook up the super drive herself, and when it didn't work, she came here for help. I'd say she was really interested in finding you. Going out of her way to help you, Steve, is quite a difference from the Sandra as I know her."

"Do you?"

"Say! What is the matter with you? 'Has she?' 'Was she?' 'Would she?' 'Did she?' is that the best you can do?"

"Look, John, how long ago was that?"

"About a week or so."

"What did she do, exactly."

"She came here and told us that you've been a month or six weeks overdue on that trip to Sirius. She wanted the drive fixed so that she could go out and look for you. I offered to go along, but she said no. So we fixed her drive and she took off like the devil was in her hair."

"Mac, you're a sucker!"

"Oh, now look—"

"So she's changed, has she? Full of remorse. Sputtering like a leaky alphasat field because she was hamstrung without a drive. Her heart was reeking with love for me, and she wanted, if she couldn't have me, to go out into the deep, unknown void of interstellar space and die where I had died, so we could be together in that last, long resting place."

"What are—"

"So John, please, for the small help I was to you, and for the love of Steve that lies within both of us, give me the drive so that I may go forth and seek he whom I crave. I want so little, John, and Steve is such a fine fellow—"

"Say! Have I been took?"

"The proper word is 'Taken' and the answer is in the affirmative."

"I'll be damned."

"You probably will," smiled Hammond. "Mac, all that dame wanted was to be the first human being to set foot on another, extra-solarian planet! She wanted to be known as the first person to ever seek another star."

"I take it that you haven't been further than a long stone's throw?"

"Shucks. I haven't even been out to the Los Angeles city limits."

"Darn her hide!"

"Yeah. I've been looking for her—and I'm as big a dope as you. I wanted to offer her the chance to pilot the *Haywire Queen* out there. I couldn't find her in the inner system and so I was going to take a squint at Pluto. I stopped off to ask if you'd care to take the run with me."

"You know I would."

"Well, that takes care of both answers. Drake is on her way—shucks, she's there already—and the second part is you—and you want to go."

"I'll ask Enid," said McBride. "Come on, we'll go right down and see her now."

Enid McBride smiled. "His asking me is a matter of form," she told Hammond. "Naturally he'll go. I think it will be swell for him to go. He needs a vacation anyway."

"But—"

"No buts. You'll go and like it. I wouldn't want you to miss anything like this for the world."

"How about you?"

Enid smiled again. "I'm no pioneer type, John. You know that. I'd be out of place—and what would John Junior do? Oh, we could leave him with Anna, if I wanted to go, but somehow this is as far as I care to get from home—my folk's home, I mean. It's funny how after seven years a woman still speaks of her parents' home as her home in spite of the fact that she has a home and family of her own."

"What'll you do?"

"I'm going to take this opportunity to go home—my parents' home, I mean. You see, Steve, Dad and John talk different languages. Dad is a metal broker on Pluto. The only reason why he tolerated John at all was because John's

lens kept Dad in business. Dad wouldn't know a cupralum pig from an acceleration cushion, though he deals in a million tons of the stuff every year. It's all on paper. On the other hand, John wouldn't know how to sell the stuff, but he sure can make it do tricks. So they sit and glare at one another and each one wonders how the other makes a living. Dad's money is obvious, and John's success is equally well-known, but how and why are lost on each other.

"So I keep 'em as far apart as I can."

"I get it," smiled Hammond. "Pretty bad, hey?"

Enid laughed, "This ring is pure iridium. Dad was horrified because he first thought that iridium was radioactive like radium and that I'd get burned or worse. Then he found out it wasn't—and offered to buy a real, honest-to-goodness platinum ring if John couldn't afford it. Then he discovered that iridium is so rare that they do not have a market price per gram and that was all right, but he also confused it with iodine, and worried about its chemical action on my hand. Poor Dad still is not sure about it, so he has to inspect it every time he sees it to ascertain whether or not it is turning green, or my finger is falling off, or that it hasn't sublimed and disappeared. You can't detect the wearing, so Dad then accuses John of either buying a new one every time I come home or making me keep it in a safe while I'm here."

"Cupralum, to Enid's father, is something that he shunts around by signing papers and which, if he shunts fast enough, will increase his bank account, though if the other guy shunts faster, will cause him no end of deficit. Space, to him, is something that you can't breathe, and the stars are little bits of brightness that twinkle on a clear night. Oh, we get along," smiled McBride. "After all, he's Grandpa now, and John Junior is likely to get a slab of Cupralum. Preferred, for his birthday. The kid'll prefer something he can chew on, I'll bet."

"So that's neither here nor there," said Enid. "You take your space hop, and I'll take Little Johnny to Pluto to see his grandparents. Frankly, Steve, I've been wondering just what excuse I could use to run off alone for a month. This makes it perfect."

"We'll stop at Hell on the way back and pick you up," said McBride.

"Fine. How soon are you leaving?"

Hammond said: "Anytime he's ready. How soon can you cut loose from the lens, John?"

"Give me an hour to get things cleaned up and I'll be on the beam."

"Right."

"I'll pack you a bag," said Enid. "Have any preferences?"

"Shirts, shoes, socks, and shaving kit, mostly."

"Want your dinner clothing?"

"Oh sure. And pack my swimming suit, too. Also my tennis racket, and see that the golf bag has plenty of spare balls. Have Timmy wax the skis and sharpen my skates, and I'll also take along the shotgun, a pup tent, the oil stove, a fur coat, a quart of whiskey, six lemons, an orange, a lime, and a bottle of Angostura. Might pack me a light lunch, too."

"Don't bother, Enid. We've got most of that stuff with us," laughed Hammond.

"All right," chuckled Enid. "He'll get one shirt and a bar of soap; one pair of socks, and a bar of soap; and so on—with a bar of soap. Well, keep 'em coasting, Steve, and see that he doesn't run off with any red-headed witches."

"If we see any, I'll bring 'em back for me," laughed Steve. "See you later."

McBride was not as abrupt as he sounded. His business clean-up consisted of dictating a letter, putting all things in the hands of his chief assistant. The rest of the time he spent with Enid, saying good-by. Whatever transpired, whatever they discussed, whatever plans they made—and they must have talked of many things and made many plans, for in spite of the familiarity of running all over the solar system, this was a big step, indeed, since for the first time in history, man and wife would be light-years apart—they did it well enough in private so that their parting was simple and quick.

John kissed Enid adequately, and said: "Stay healthy."

Enid laughed and said: "Stay whole!"

And then McBride was in the *Haywire Queen* and the air lock was cracked. The big ship lifted gently and zipped out of the lock with a casual disregard for distances. Unlike Drake's precision take-off, the *Haywire Queen* went through the open door with the air of wanting to leave quickly because there

were better things to do than worry about hitting the center plus or minus an inch.



Enid pointed out the Dog Star to John McBride, Junior. "That's where your daddy is going," she told him. Junior McBride was more interested in the teething bone that he had clamped between toothless gums, than he was in the stellar regions.

He knew his daddy would be back.

The *Haywire Queen* approached and passed the speed of light from the hard side, and her terrific velocity dropped down to a figure that was expressible in miles per second without running out of zeroes. Below, and thirty degrees from the axis of the ship, Sirius and the Dark Companion beckoned from less than a thousand million miles. The lower dome of the ship sported the faces of the men, who were laying on their stomachs, looking down at the splendor of the first binary ever seen by man. Hammond mentioned it, as a matter of fact.

"How about Drake?" asked McBride.

"We're still the first *men*," returned Hammond.

"Wouldn't Drake howl to hear you say that," laughed McBride. "She's been suffering under the fact that every time she did anything new, she had to qualify it by saying: 'The first woman—' Well, she's got something this time."

"Think it'll satisfy her?"

"Not until someone proves definitely that Thomas Edison, Franklin Roosevelt, William Shakespeare, George Washington, Richard the First, Julius Caesar, and Jack Frost were all women."

"Well, let's get the hemis working. We'll never know whether Sirius has planets until we do. I'd hate to sit in the *Queen* and go through all the growing pains of looking for planets by observation."

"Yeah, that would take years. What's our velocity, Larry?"

Timkins looked at the velocimeter; squinted through the instrument quickly, adjusting the thumb-screw; and then said: "Thirty-four thousand and dropping at one hundred feet per second, per second, per second."

"We can get good pix of anything close enough to the primary to support life—also big enough, too—in about thirty minutes exposure," said Hammond. "We'll take two shots in each direction, since I've got six hemispherical cameras. That'll give us complete overlapping coverage and double protection against dust streaks. Let's go. Also cut the drive by half."

For thirty minutes the ship plunged on through the Sirian system at the double deceleration. Then for fifteen minutes, the entire personnel was in the darkroom, waiting for the first glimmer of the plates. And at the time that the plates were finished, the velocity of the *Haywire Queen* had dropped from thirty thousand-odd miles per second to velocities normally used in mere interplanetary travel.

The super drive was cut and the ship coasted under standard drive at thirty feet per second, per second, acceleration, and the men hung the plates up in the darkroom and began to inspect them for telltale streaks.

"Here's one," said McBride. "About four hundred million miles from Sirius."

"And another," offered Larry, plying dividers and log tables, "about three thousand million."

"Got another," offered Hammond, "but it's doubtful as a possible landing place. Almost ten thousand million miles from the primary. Bet it's colder than a pawn-broker's heart."

"Couple more on my plate," said McBride. He went to the formerly empty solar map and added the discoveries according to scale. "But that one at four hundred million is my best bet."

"Sounds reasonable," agreed Hammond. "Sirius would support humanoid life at that distance. Let's concentrate on it."

"Good. It's in fine position to be concentrated on. Let's see, now, what should we be looking out for?"

"Might be seetee matter," suggested Larry.

"Good. How do we find out?"

"We don't until the last ditch. But it is the most important, nevertheless. We wait until everything else has been disposed of and then make for the planet. Just outside of the atmosphere we heave 'em a rock or two and watch what happens."

"A slow moving rock?" grinned McBride.

"Doesn't really matter. If it is slow enough to keep from friction-incandescence, fine. But the eruption made by seetee contact is quite a bit different, spectroscopically. Also we can check the explosion with counters.

The by-products of such a bit of eruption is full of nuclear radiations. Mere incandescence is just that and nothing more."

"Well, that's that. We can wait. What's next?"

"Radioactivity. How much and what kind? Atmosphere. How much and what kind? Et cetera. Also how much and what kind? Do we intend to land?"

"I don't know. After all, we came for the express purpose of trying out our drive on an interstellar basis, you know. It can be done with ease, neatness, and dispatch. Seems to me that a landing on one of those planets will have to be made attractive or we won't. We're equipped for all kinds of spacial research, power research, and so on. But we're not equipped for much planetary investigation, exploration, or diplomatically involved intrigue."

"Going to let Drake get away with being the only person making the first landing on an alien star system?"

"I don't give a care what happens to Drake. She can come busting in with the safety valve tied down if she wants to. Some day she'll learn that sticking that pretty little snoot of hers into strange places is a fine way to have it knocked right off of the front of her face. We're interested in technicalities, not in getting involved in a storybook adventure. Meanwhile, let's take it strictly on the easy side and investigate everything from the solar radiation from Sirius to the secondary radiation produced by Sirian radiation in the super-stratosphere."

Larry began to fiddle with the radio. There was nothing on the electronic radio at all, and Larry said: "Well, didn't expect it, really. No culture worthy of the name would be using radio in space. Too inefficient. And if they got off of their planets, they'd be using gravitics." He turned to the space radio, and covered the communication bands of the electrogravitic spectrum, switching from band to band quickly. Halfway across the third band, the panoramic tuner came to a definite stop and retraced itself minutely, vacillating a bit until the signal came in clear and clean.

"What happened to Drake?" asked Timkins. "Listen. Here she is."

The gravitic radio was calling: "*—Haywire Queen. Calling Haywire Queen.* This is Sandra Drake calling the *Haywire Queen*. This is an automatic transmission set for break-in. As soon as this call gets to you, answer please. The answer will register here and we will be able to make this two-way. This is Sandra Drake—"

"Uh-huh," said Hammond, turning down the gain to a reasonable level. "Larry, shoot her an answer."

Timkins snapped on the transmitter, tuned it to the same band, and said: "This is the *Haywire Queen* calling Sandra Drake. *Haywire Queen* answering Drake. Come in, Sandra Drake. Answer."

They listened to the automatic broadcast for some minutes, and then in the middle of a sentence—"This is Sandra Drake calling the *Haywire Queen—*" *Click*. "Hello, fellows. Got here finally, didn't you? Glad to have you come in. What's new?"

Hammond took the mike. "Hello, Sandra," he answered. "Nothing new. Where are you?"

"On planet number five. That is the one that I think is somewhere about five hundred million miles from Sirius. Know it?"

"We think so. It's dead ahead. Yeah, wait a minute. Larry has a directional bearing on you and it is the one we're approaching. That takes care of that."

"Well, come on in and I'll build you a cup of tea."

"You find everything all right?"

"Everything's perfect. Only thing, they would like to have someone here that knows all about the gravitics. They're not too sharp. Frankly, neither am I, so you're the guys who'll have to do it."

"You've been there quite a bit," said Hammond. "How's conditions?"

"Pretty good. Air is O.K., though slightly pungent in smell. The people are very much like humans, though they have their big differences which take them out of the human class."

"For instance?"

"Well, they are all covered with a funny kind of hair. It's a sort of half-hair, half-feathers kind of stuff. It's as soft as a baby's scalp and on a dog or

something like that it would be beautiful. I'd like a coat made of it, frankly."

"I'll bet they appreciate your offer to wear one of 'em for a winter coat," said Hammond dryly. "You haven't changed a bit, have you, Drake?"

"Oh, I wouldn't say that," said Sandra. "After all, I was merely trying to explain the beauty of their skin."

"You gave yourself away," said Steve Hammond. "Like as usual, Sandra Drake thinks of everything in accordance with how it will couple to her, or her name, or her reputation."

"Now, you're being hard," complained Sandra. "Give me a break, Steve. You shouldn't take issue with me for a statement of that kind. After all, it was just a sort of slip of the tongue. I'm not really thinking of skinning one of them for my coat."

"If I were you," put in McBride, "I'd think hard of one other thing that might be closer to home. D'jever think that you are in no position to do any skin collecting? The odds are agin' it. But, Sister Drake, those birds are! You might enhance the beauty of one of their females some day. How would the pelt of Sandra Drake look on the living room floor, nine light-years from Terra? Take it clean and easy, Drake, or you might not get back to Terra with that satiny, soft, practically flawless hide of yours intact."

"What do you mean, 'practically flawless'?" snapped Sandra.

"Well," drawled McBride, "I've never seen all of it."

"Why don't you give me the benefit of the doubt?"

"I wouldn't give you any benefit of any doubt," McBride told her. "You're probably concealing something."

"Why—" the radio broke down into a series of liquid, spluttering sounds as Sandra strove to keep that throaty contralto from sounding like a fishmonger's.

"Whistle," chuckled Timkins. "Then count ten. Then let's get back to the problem of the Sirians."

"Take it, Sandra," laughed Hammond. "We were only kidding you. Or—can't you take it?"

The spluttering died, and then that throaty laugh came back again. It was slightly forced and they knew it. The chances are that Sandra knew they knew it, but she didn't want to give them any more reason for laughter at her expense. Then she spoke, directly and honestly, both factors due to the fact that she was sure of herself and now could afford to laugh at them.

"Well, stop worrying about Sandra's hide," she told them. "This gang down here are fine people except that they can't talk Terran. They'll do anything for me that I can make them understand. That's the trouble—getting them to understand. But that's coming. I'm teaching them to speak Terran. That should fix things up fine."

"Why not learn to speak Sirian?" asked McBride.

"Why? Let them do the work. Learning a new language is not Drake's idea of a year's fun."

"O.K., sister," grinned Hammond, winking at McBride. "But you'll find out that there is something to those old adages. I'm thinking of the one that begins: 'When in Rome, et cetera.' Those old boys used to dust off some old saws, but there is a lot of meat on them."

"And contradictions. No, fellows, Sandra doesn't like talking in something that sounds like a phonograph record played backwards. Besides, these fellows have a pretty sharp capacity for understanding. I've been here for a week or so, and already they can understand a lot of what I say. Frankly, better than I could."

"Play it your way, then," said McBride. "But look, you say they're nice guys?"

"Sure. When I landed, they gave me the old send-off. I was taken to the royal house and given the prize suite. I'm given everything, as I said before. They look upon me as the guy who'll give their world the benefit of the Terran and Solarian scientific achievements. That's not true, of course. It'll be fellows like yourselves who really understand it. But nevertheless, I'm the harbinger of spring. I'm the guy who pointed the way for the rest of Sol's children."

"The Moses in the bulrushes?"

"Sort of like. I'm just lucky, and I know it. If I'd come second, they wouldn't pay any attention to me at all. But since I came first and now that I'm talking to my friends, they will obviously think that I'm calling for them to come

and help them ... their world's name is Telfu, by the way ... Telfans out of their scientific rut. They have the glimmerings of the gravitic spectra, but it's like the difference between the Leyden Jar and the electron microscope. It'd take a hundred years before they got off of Telfu if we hadn't got here first."

"If they're really O.K.," said McBride, "we'll help."

"Thanks," said Sandra simply. "That'll be for me, too, you know."

"Yes?"

"Sure. They'll thank me for coming first, even though they know I'm not the bright guy with the answers under my skull. I've got a good thing here, and I know all of you well enough to know that you won't spoil it."

"No?"

"Sure you won't. After all, there isn't one of you that would care a rap for what they have to offer in the way of historic gain. The old moola, sure; and there's plenty of it to be had for all of us. You'll go down in their histories as the geniuses that gave them a boot in the tail worth a hundred years of solid research. I, and I'm sure you'll permit me, will ride in on the tail of your coat."

"O.K. Well, we'll come in. But not for long this time. After all, we're interested in tinkering with the new drive, not making diplomatic overtures to a bunch of aliens. We'll leave the latter for the Solarian Government."

"How soon'll you be landing?"

"Not too sudden," said Hammond. "We're going to make a few space-checks first. We're getting cautious in our old age."

"Shucks," said Sandra disparagingly, "there's nothing to it at all."

"Well, could be, but we'll run this show our way. There is no objection to your leaving?"

"No. Definitely not. They'd be sorry to see me go, but it is personal affection and the possibility for their ultimate gain that makes it so. They wouldn't dare detain me even though they might consider it. To my knowledge, they haven't even considered it."

"Why wouldn't they dare?" asked McBride.

"Afraid. After all, they know that both of us came from a star nine light-years away. They haven't even got the primary drive, let alone the third-derivative drive. Any untoward move to a Solarian would bring the devil himself down about their ears and they know it."

"I suppose so. We could drop plenty of stuff on 'em with a half dozen space cans. And a couple of monolobar mechano-gravitics would scramble up the works of any fleet of stratosphere planes they could send against us. Never gave the gravitic armament much thought, but it could be done. O.K., Sandra, as soon as we sniff the air and check our gas and water, we'll be in."

"I'm going back to bed, then," said Sandra. "Slip me another call before you land and I'll have the village band out to meet you. That's a promise."

Steve Hammond turned to McBride after Sandra had clicked her transmitter off, and said: "No use checking for seetee matter, is there? Seems to me that Drake would have found it out the hard way."

"No, we can skip the seetee. But Drake may not worry about radioactivity but we will. We'll check for it; I'd like for John Jr. to have a brother or sister some day—with the proper amount of arms, legs, fingers, toes, ears, eyes, noses—"

"What's the proper amount of noses for a son?" asked Hammond.

"One," grinned McBride.

"A kid with two noses could smell a lot," observed Timkins.

"*Phew!*" said McBride holding his nose. "That was fierce. Man the counter and check the region for hot stuff, Larry. Looks like the landing of LaDrake saves us a lot of work. The physical properties of ... Telfu ... seem to be all right. So we'll go to work on the electrical properties, the nuclear properties, and also see if there's anything running around loose in the gravitics other than the inherent mechanogravitic property of matter."

Larry Timkins set up a series of plungers on the control board and locked the pre-set operations into the autopilot. "This," he said, "will hang us on a

logarithmic spiral approaching Telfu. While we're roaming around the planet, we'll check the hot-properties of the neighborhood. Any comment?"

"Nope. Give 'em the works."

Timkins drove the coupler button home and the *Haywire Queen* swung gently to follow the pre-determined course.

"You know, Steve, there's a cod-liver-oil smell about this, somewhere."

"So? What's fishy?"

"The old tub isn't behaving like a lady."

"What do you mean?"

"There's a big drop in efficiency compared to when we left the Plutonian Lens."

"How much?"

"Not too much. But it's getting progressively worse."

"Y'don't suppose we've hit upon some saturation factor in the secondary drive?"

"I'm not saying. What do we know about it? What does it work on?"

"Glibly speaking, it works on the inherent qualities of space. We wrap ourselves up in a space-warp of sorts, and then shoot out a couple of hooks that catch on to the gravitic-propagational continuum that permits the planetary masses to exert Newton's Law of Universal Gravitation. It has been called 'sub-ether' but that is like multiplying with unreal numbers. After all, the 'ether' has never been defined, isolated, explained, or held in one hand. If the prime 'ether' has never been satisfactorily established, we shouldn't go on building our houses on a foundation that doesn't have any sound basis."

"Both electronic and gravitic spectra must rely upon something for propagation," objected McBride. "For lack of taking it apart, brick by brick, and feeling each stone, let's continue to call them 'ether' and 'sub-ether.'"

"O.K., sport. But to get back to the drive. Have we got a saturation point? Or some sort of gravitic fatigue? Either of these would be indicated by a gradual decrease in efficiency."

"Larry, set up a sigma recorder and let's see if we can check the curve of inefficiency. It's getting worse, you say?"

"Apparently. I didn't notice it before. But it is quite apparent now. Must be non-linear, because if this falling-off had been linear, I would have noticed it long before this. An increasing curve would not be noticeable until a sufficient interval had been passed for it to become evident. Yeah, I'll slap a sigma recorder on him and see what makes."

"Meanwhile, let's get busy with the detectors."

The counters clicked for a few minutes, and McBride finally reported that Telfu was no higher than Terra in radioactivity. Hammond established the intrinsic electronic charge on Telfu as being only a few million volts negative with respect to Terra.

"Not enough to worry about," he said. "The first touch with the stratosphere layers will take care of that without a glimmer. Wouldn't dare without an atmosphere, but we have plenty of air to cushion the charge and let it leak off in the upper layers where it is ionized by Sirius' radiations. What's with the gravitics?"

"Bit of something in the electrogravitic. Can't place it. Not enough to worry about."

"What is it like?"

"Well, it is not E-grav radiation. It's a sort of dip, or valley, in the radiation-pattern of this part of space. A place where the normal density of E-grav is less."

"How much?"

"You tell me. The free-running gravitons are never high enough to do more than flicker the finest instrument. The threshold is way, way, way, way down in the mud. So here's a place where we have less."

"Sort of like having nothing and wanting to share it with someone?"

"Not much better. Oh well, a lack of free E-grav energy surely isn't anything to write home about. Might be a factor of the Sirian Double. After all, who knows what kind of effect that little, dark-red, dense-as-hell devil will do to gravitic threshold levels."

"So it's a safe bet—"

Timkins came running in, waving a sheet of cross-ruled paper. "Hell's bells," he yelled. "We're it! Our drive is approaching zero efficiency as the third power of—"

Above, in the working innards of the *Haywire Queen*, great circuit breakers crashed open. Smaller switches added to the din as they clicked open, one after the other. Pilot lights on the polished black panel began to glow an angry red and alarm bells created such a din that speech became almost impossible.

The drive went off.

And the men and their portable equipment left the solid floor and began to float aimlessly across the room in midair.

Hammond clutched wildly at a spectrograph, and caught it.

"Catch!" he yelled at McBride, hurling the heavy instrument at John.

McBride folded himself over the instrument with a grunt of escaping breath. The act did two things. It sent Hammond across the room to the emergency panel in one direction and McBride went in the opposite direction to the navigator's calculating machine. McBride caught the navigator's table at the same time that Hammond caught the emergency panel.

Steve fought with the emergency panel and succeeded in setting up about eleven feet per second deceleration. McBride lowered the spectrograph to the table and seated himself in the chair.

"Woah, Nellie," grunted McBride as the alarm bells ceased. "Where do we go from here and how fast?"

"I dunno, but we're leaving both Sirius and Sol at a terrific velocity and a deceleration of eleven feet per. From a mental calculation of the fundamental drive at this velocity, I'd say it would take about fourteen years to get down to a stop."

"What happened to the emergency relays?"

"They worked," said Steve dryly. "Yeah, they worked. But the inefficiency extends to the fundamental drive, too, it seems. I'm beginning to think that this is not inherent."

"That's a quick decision."

"Sure. But the prime drive is O.K. The meters say so. It's just inefficient as the devil which is not true of a good drive. Holy smoke! We're getting efficient again!"

Timkins picked himself off of the floor painfully. "Uh-huh," he grunted. "Also, we're leaving Telfu behind at a fierce rate. Can you keep that eleven feet prime acceleration for a bit?"

"We're going to."

"I'm going to dash madly upstairs and hang the sigma recorder on again. Something is slippery here."

"What's our velocity at the present time?" asked McBride.

"Up in the fifteen thousand miles per second," answered Hammond.

"Hm-m-m. Then at what point with respect to Telfu did the drive go out?"

"About a million and a half miles, roughly."

"A minute and forty seconds from spot to conjunction," mused McBride. "If, little playmate, we can get power again after one more minute and thirty seconds-odd, we'll feel more or less sure that it is Telfu and not us. Larry!" he yelled. "Any sign of upswing?"

"Yup," said Larry. "Sure thing!"

"Set the super drive up on test power with automatics to turn it on as soon as the overload point is passed," said McBride. "We won't blow any fuses with test power."

Hammond hit the test buttons and then settled down to wait. Then the drive cut in again, and they all slid down in their chairs.

McBride grinned. "They must not like us."

"Something must not," laughed Hammond shakily.

"Telfu?" asked Timkins entering with the last sigma curve.

"What does it say?"

"We passed through a negative peak. We hit a new low in efficiency at conjunction with Telfu."

"How much?"

"Less than a half percent."

"Jeepers. That is a new low in gravitics. Can we think our way out of this one?"

"Why?"

"As much as I dislike seeing Drake, I'd not force her to live on an alien planet. I'd feel better at marooning her for a couple of years if I knew we could go in and get her."

McBride laughed. "Got to have the last laugh, hey?"

"Meaning?"

"Marooning her wouldn't be half so much fun if it is impossible to get her out. Marooning her when we have the means to get her out puts it strictly in our own lap. Right?"

"I suppose so. We could laugh at her honestly then."

"She's strictly a stinker," agreed McBride. "I get that cod-liver-oil smell now. All that soft soap and palaver she was handing out about our being the boys with the brains. We were the guys who would be responsible for lifting a struggling civilization up from the primordial slime by our brain and our genius. Baloney!"

"I get it," growled Hammond. "She's stuck. God knows how she landed—probably emergency and shot her load of battery juice. Anyway, she could land under emergency battery, but taking off is a megawatt of another color, battery-wise. They aren't equipped to make a take-off. Idea being the old one—don't start if you can't stop."

"She's a bright girl in her own stinking way," said McBride. "She's been around this gang long enough to know that if a way is possible, we'll think of

it. Oh, sure, that's a brag but we've done pretty well so far. So inveigle us into the same trap she's in and then ride out with us. She'd roast in the brimstone of the nether regions before she'd wail for help honestly. But if we get stuck with her she's got two outs. One, we may be able to think our way out. Two, at least we are Terrans like she is."

"Meaning?" asked Hammond darkly.

"Frankly, Sandra Drake is an awful lot of woman, and she knows it. She'd make a plaster saint turn to whistle at her if she turned on the old charm. And with no competition, we'd be fighting one another for the privilege of polishing her shoes."

"Fine future."

"No thanks."

"I'll have a bit of that, too. Well, how can we slip her the old triple-cross?"

"Steve, you'd throw a woman to the lions?"

"With that woman, I'd hate to do it. The S.P.C.A, would haul me in to court for subjecting poor, dumb, defenseless lions to cruelty and inhuman tortures. You're darned right I'd heave her into the drink. But I want to do it in such a way that Sandra Drake will know that it was far from purely coincidental."

"O.K., Steve. We're with you. Larry, throw the *Haywire Queen* into an orbit around Telfu just outside of the danger zone and slap another recorder on the drive. Make it a high velocity orbit, powered all the way. We should be able to circle Telfu in about fifteen minutes with the super drive. Check?"

"Sure. Here we go."

"Meanwhile, Steve, we'll check a few items on the drive itself. I'm beginning to suspect a huge and celestial soak-up of gravitic power in the region of Telfu."

"We can set up the small, experimental drive-model complete with power recorders, spring balances, and torque measuring devices and work on that."

"Swell. That's the ticket. Let's go."

Hammond hauled the model from the cabinet and plugged in a complex cable from the master control panel. He juggled the dials until the gadget started to work, and then they began to check the efficiency of the device.



McBride muttered: "Power generating equipment is running O.K."

"Yeah," agreed Hammond. "Everything's on the beam from the explosion chamber to the inverted alphasat. We've got plenty of potential power handy. Larry, zoop in close and check the power equipment on a pure, resistive load."

"You mean shut off the drive and coast through the zero region with no drive and with the gravitron running at full output on resistance load?"

"Right. This fishy smell has a rare odor. I think we're on the trail of it."

"O.K., Steve. Can you wait about three minutes? The first encirclement of Telfu will be over then and we'll have our first experimental curve."

"We'll wait."

The sigma curve was completed, and Larry circled far out and made a fast run toward the planet, in a course similar to the one they used on their first try.

Meanwhile, Hammond looked at the curve and grinned.

McBride looked over his shoulder and grinned, too.

Hammond slapped the curve down on a drawing board and began to plot efficiency against a polar co-ordinate. The curve was roughly circular, but exhibited a tendency towards a cardioid. McBride played with the figures for a minute, and as he opened his mouth to say something, the *Haywire Queen* gave that sickening lurch and changed abruptly from super drive to the emergencies.

"Darn!" said McBride. "This everlasting acceleration changing business is going to make a nervous wreck of me yet."

"Also physical if it is taken in too large doses," grinned Steve. "The human anatomy can accept velocity without limit—well, up to the point where the ultimate velocity is reached. We've gone a goodly hunk of stuff over the speed of light."

"That's questionable."

"We came over from Terra in a lot less time than light. That'll be arriving nine years from now."

"Uh-huh. But don't forget we wrapped ourselves in a space-warp and ran the space-warp. I think that we can safely assume that the warp is another space and that we were not traveling better than the speed of light with respect to our own space."

"Whoof! What a theory! Drag that one past again, slow enough so I can climb aboard."

"You got it," laughed McBride. "And if it smells, you fling out a better one for us to shoot holes in."

"O.K. But to get back to velocity, the human anatomy can stand velocity without limit. Period. Argue if you like, Mac, but that's my statement. No one has ever been able to prove that velocity alone is harmful to man, beast, bird, or fish!"

"I'm as silent as the tomb."

"Acceleration can be adapted to—in meagre doses. A man can stand up under 4-G. On his tummy, lying down, 8- or 9-G isn't too hard on him. Dunk him up to the breathing-vents in a good grade of oxidized hydrogen and 15-G is possible without too much harm."

"Yes. O Learned Scholar."

"But, students," said Hammond standing up and taking a bow. He was interrupted by the resumption of the super drive which, being set at ninety feet per second per second apparent instead of eleven feet, caught him off balance and almost dropped him on the end of his nose.

"What I was saying," laughed McBride, "was the effect that rates of change of acceleration have upon the anatomy."

"As I demonstrated," grinned Hammond from the floor, "it is changes in acceleration that cause havoc. It causes jerks—"

"To sit on the floor," chuckled McBride. "Get up. Stop playing on the floor, Steve, and take a squint at this curve. Plotting an exponential factor for the ordinates of the graph, using Telfu for the center, we find a locus of equal power-soak-up out here—which I estimate to be a little more than two hundred thousand miles!"

"Ah, the wonders of analyst," said Hammond. "With a defunct drive and a wild idea, Jawn McBride hauls a satellite out of the sky and plants it—Here!"

"What do you think?"

"Who am I to argue with people who understand the mysteries of A to the Xth power equals zero, divided by the date of the month times the ace of spades, equals eleven o'clock. All joking aside, Mac, it looks right to my uninitiated mind."

"Does, hey?"

"Sure. That means that said moonlet—I say moonlet because our pix show that Telfu hasn't anything worthy of the name of a full, honest moon—must be high in cupralum."

"Sort of hard to believe."

"Yeah, but not impossible. It's quite believable that the right alloys should be found *au naturel*, so to speak. There's nothing tricky about cupralum. Mix it together and smelt it down—*voila!*—cupralum. A totally useless and good-for-nothing alloy prior to the discovery of the gravitic spectrum."

"Must be fairly large," suggested Timkins.

"Sure—according to man-made standards. Celestially, it might be a mere scrap of dirt. A sub-sub-sub-microscopic bit of cosmic dust less than a hundred miles in diameter."

"Ugh," grunted Larry. "You make man and his works sort of insignificant."

"We are. Do the planets care what we do on their miles-thick hides? Do the suns care that we wonder at them? Does the cosmos give a rap that we chase from planet to planet and from sun to sun?"

"You make it sound as though they are capable of thinking."

"If they did, we wouldn't know about it; and they wouldn't know we existed. Proportionally, man is smaller than the filterable virus. So we have a slab of cupralum, which is—according to Mac—Here! That's fine. It blankets Telfu like a complete shroud, as far as the good old gravitics go."

Larry Timkins looked up from a page of scrawled equations. "A slab of cupralum a hundred miles in diameter, rotating in the mechanogravitic field thrown out by Sirius would certainly soak up every bit of power. Must be a slick tie-in. The gravitron puts our O.K. on a resistive load. Hooked to the drive, everything goes *phhht*."

"Sure. That's part of the trouble. It's the drive, coupled with the general gravitic interference cut up by Soaky."

"Soaky?"

"I have hung a name on the satellite. Heretofore it has been nameless. We have named it *Soaky*."

"There is a slight discrepancy between this cardioid and the calculated curve," said McBride. "Obviously, the cusp would be on a line between Telfu and Soaky, projected from the satellite through the planet to the far side. We orbited around the planet and were closer to Soaky on the side he was on—"

"Is that syllogistic reasoning?" asked Hammond. "Or sheer conjecture? How about shadow?"

"This is quite a wide effect."

"Any shading of Soaky's sphere of influence would tend to deepen the cusp like that. That cardioid is such a curve; there's no reason to doubt that Telfu would tend to shade the field."

"Larry. Can you calculate the field absorption of a standard model planet with the above figures?"

"The attenuation?"

"Yes."

"Sure. It'd help if I knew the chemical components, mass, physical constants, electrical properties, gravitic properties, and nuclear emanations. How close do you want it?"

"Plus or minus twenty percent."

"I can give that to you without calculating," said Timkins. "Telfu is similar to Terra within twenty percent. Terra's attenuation amounts to twenty-nine percent; in other words, the attenuation due to the presence of Terra in the light-line between source and measuring device is twenty-nine percent greater than it would be if Terra were not there and the spacial attenuation only cut the strength."

"Thirty percent, roughly, because it's easier to figure," said McBride. He made calculations, set them down linearly as to the magnitudes, and then transferred the vectors to the curve.

"That's one large bit closer," he said. "We'll better that, some day. But for now, playmates, I've had my Idea-for-the-Week. Let's cut us another caper around Telfu at right angles to this curve. One side will pass the peak and the

opposite side will cut the cusp. Same distance, same speed, same everything. Follow?"

"At some distance."

"I believe that we will find a place where the cusp really comes down closer to Telfu," said McBride. "How much drive inefficiency can we tolerate and still lift?"

"From Telfu? Not enough to keep the breakers from blowing. And don't say wire 'em shut. They're right on the ragged edge now, on account of we know what we're doing and do not want to blow circuit breakers during experiments unless they are really in trouble. But the gravitron-cupralum driving equipment is not our only ace in the bucket. The emergency batteries, though inefficient, can still put us down and get us off. Providing, of course, that your map there gives us a chance."

"Not knowing the orbital constants of Soaky; the plane of Soaky's ecliptic: the rotational features of Telfu, we are taking chances. One rotation of Telfu might be plenty safe if we hit it on the nose. Two might put us out here and then we'd have to go through seven years of astronomical investigations before we found the place where that cusp came in again—and we'd probably have to wait anything from sixteen to nine thousand years before Soaky passed overhead again. The latter might get boring. But we can take a chance on one day, plus whatever angular movement Soaky makes with Telfu as center."

"Think Soaky's ecliptic is fairly close to Telfu's equator?"

"Within twenty or thirty degrees. I'm assuming the old theory of the Planitesimal Hypothesis. Sling out your molten stuff, let it condense, and you'll find everything rotating in the same direction in about the same plane. Might be clockwise or counter-clockwise, but only one way per solar system. One moon in all of the junk that goes around Sol is contrariwise—and they think that was a captured wanderer. The greatest obliquity is somewhere near forty degrees, most of the large planets being less than ten, I think."

"Celestially, I believe it may be impossible for a satellite to hold an orbit whose plane is vertical to the planet's orbit. I've never given it any thought,

but it sounds dangerous to the satellite. Also, Sirius' tidal drag would tend to bring all the planets' axes into vertical line, too."

"Oh the devil. I want to land. If waiting overnight is dangerous, we'll slide in there and out again inside of an hour. But, darn it, I want to plant my number eleven EE's on that planet. Anyone agree?"

"Anyone who doesn't like the idea may get out and walk," said Hammond. "Hold your hat, fellows. Here we go again—"

Sandra Drake reached out of her luxurious bed and pulled a cord. She did it in a languorous move, like a lithe and lazy cat. She did it with a sort of God-given right to do so, and her expression was one of deep self-delight. Whatever she got from Telfu, they owed to Sandra Drake—

Her second pull on the call-cord was more of an impertinent yank. Her self-delight changed to exasperation that they should keep her waiting. Yet she would forgive them, for they were ignorant, in forgiving them her grace would be more evident. They would love her the more for forgiving them their sins of omission—

Sandra's third pull caused the collapse of the call-bell box, and the cord fell, landing in long, graceful loops over her outstretched arm.

Sandra rolled out of bed and threw the cord across the room, where it draped itself about the throat of a marble nude of a Telfan woman. It could not have been placed there with more delicacy; adding just the right touch of decoration to the nude. The center of the cord depended across the chest of the statue in a graceful loop, the bottom of which crossed just above the upper pair of breasts. The ends of the cord passed once more about the throat in opposite directions, and the ends crossed the looped center to dangle between the lower breasts.

The decorative touch did not strike a responsive chord in Sandra Drake. She wanted rip-roaring action, not interior decoration. So she stamped over and jerked the cord from the statue and tried to rend it in her hands. She was not strong enough to do the cord any damage but she did succeed in breaking a one-inch fingernail.

She stormed and stamped, and said a few things that are better mentioned in the abstract, including references to the statue's maker and his family for several generations coming and going. To Sandra's Terran-minded ideas of beauty, the statue was an abomination in spite of its perfection of workmanship. It was not merely un-Terran and therefore strange, it was almost-but-not-quite human, and therefore downright repulsive, and Sandra said so in unladylike language. That the same reactions, in reverse, applied in the Telfan-Sandra relationship was not yet clear to her. Her language sounded more adapted to caisson workers, space-ship builders, or mule skinnners than it did the luxury of her present abode.

Then at long and exasperating last, the door opened gingerly and a serving woman entered.

"Well!" exploded Sandra. "Where have you been?"

The woman said something clear and articulate, which meant she was very sorry but which meant nothing to Drake. That made Drake boil merrily.

"Can't you speak Terran?" stormed Sandra.

The woman came into the room, followed by another.

"Who are you?" shouted Sandra. "Where's that other one—I can hardly tell you apart."

The first Telfan woman turned to her friend and said: "She's throwing another fit."

"She wants the Lady Thani. Thani is the only one who can speak much of her language."

"If I were Thani, I'd slip a thumb into each eye and pry."

"I wouldn't waste my time on that," returned the second woman. "I'd just make away with her and forget about it. I wouldn't care to have my sleep disturbed by blood, screams, and torture."

Sandra huffed up tall. "Will you two creatures stop gabbling at one another and get me Thani. Where is that creature?"

"Yes, she wants Thani. I heard her mention her name."

"If Thani isn't here, get me Tet'h. Or Gormal. Or Elyon."

"How can we tell her that Thani, Tet'h, Gormal, and Elyon went to meet the other Terrans?"

Sandra heard the names and the word *Terrans*. "Did they run off and leave me here?" she yelled.

They shook their heads.

"Go ... yes?" asked Sandra.

"Go ... yes!" answered Delya.

"I want to go, too."

"I ... go ... no," said Delya.

"Not you, me."

"You ... no?"

"Me ... yes."

"Me ... yes!" agreed Delya.

Sandra put both palms against her cheeks and gave vent to a yell of sheer frustration. Then she calmed once more. "Did every one of you that knows a word of Terran go?"

"Tonla, I think she's asking about Thani and the rest."

"But how can we tell her?"

"Do we want to? If all are like her—this Terra must be a bad, bad place indeed. And she is but a female. What must the males be?"

At this point it must be recorded that the first Interstellar incident was averted by Sandra Drake's refusal to work in learning the Telfan language. Drake's possible actions if she had been able to understand Delya's remark might have led to the First Interplanetary War. Amicable relations resulted from Sandra Drake's ignorance.

"After all," said Tonla, "they went because there isn't much of her language between all of them. All together they may be able to converse with the

Terrans."

"And Elyon says that she is quite uninformed as to the technicalities of this device which will not work on Telfu. She inferred that these others know much about it. They are the ones to contact if Telfu is to gain. Why shouldn't they all go?"

"Had I the right, I'd have sent them," said Tonla. "We'd better get out of here before this woman gets violent. I think she's about to start throwing things."

"She should throw a fit," sneered Delya. "Only the very beautiful can behave in that arrogant manner."

"Or the very rich."

"Name it the very desirable. Thani is very desirable, and yet she does not raise hob with Tet'h. And Thani is not only beautiful, but she is wealthy, too."

"And Tet'h is not without his own desirability," smiled Tonla. "Nor his wealth. Beauty walks in the arms of grace. She has neither."

"Let's get out. And let us hope that all Terrans are not as nasty as this one."

"I fear, though. If I were a Terran, I'd never have come to get her," said Tonla. "Unless she and they are well met."

"Perhaps they are afraid of the bad impression she'll make if they leave her here."

"You hope for that?"

"No race could be that bad."

Sandra mustered enough coherency to ask another question. "How can I get to my friends?"

Much negation.

"Can't anyone understand me?"

More gestures of complete misunderstanding.

"Get out!" yelled Sandra, and then as they started to leave, Sandra exploded again. The slamming of the door coincided with the first eruption, but the molten lava and hot ashes fell on an empty room.

"If she'd bothered to learn one word of Telfan, they'd have taken her," said Delya. "But they couldn't weigh down that little flier with one more—especially one who could be of no use to them. They'll return for her later."

"Too bad we can't put postage on her and mail her back to this Terra of hers."

"She'd come back stamped: 'Mail not wanted!'"

Sandra swore a few blood-curdlers and won her point by making an impression on the marble statue with the hard, sharp corner of a heavy metal box that stood on the table beside her bed. Then she ripped out of her pajamas and dressed quickly. She ran from her room and confronted the first man she met.

"Where are they?" she snapped.

He shook his head and pointed down the hall.

Drake followed the pointing finger to a large room. She stamped in, obviously interrupting some sort of governmental meeting.

"I want to go to my friends," she said imperiously.

The man at the head of the table shook his head sadly.

"I must go to them! Or," she asked superciliously, "are they coming here?"

More shaking of the patriarchal head.

"Can't you understand, either?" she stormed.

A shrug of the shoulder and a shake of the head gave Sandra to understand that she was speaking in an alien language to them.

"Crano!" she snapped. She didn't know its meaning, but it was the only Telfan word she knew, and she did know that it was a term signifying that the receiver of the epithet was slightly less than educated.

The elderly man went white. Two of the younger men arose, came forward, took Sandra Drake by the arms—one to each—and removed her from the chamber. They were not gentle, and on any inhabited planet employing the use of the Terran vernacular, she had been "Bounced!"

And Sandra knew it.

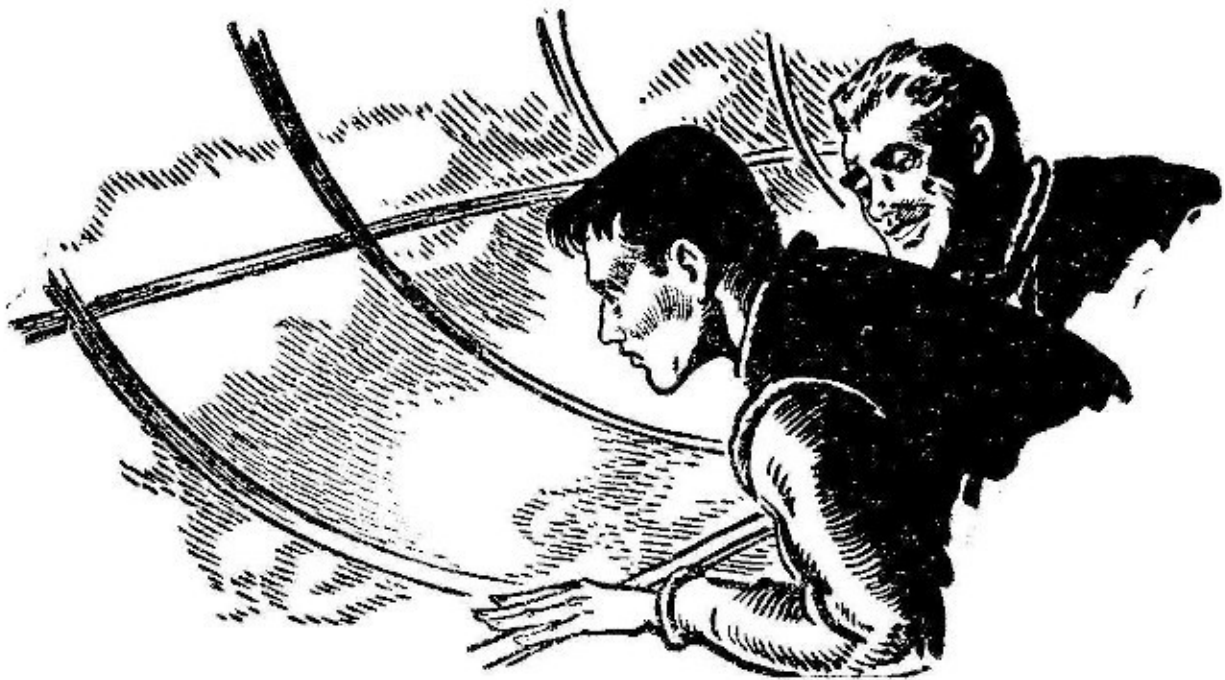
And then there came a bit of understanding. It hit hard. And in the brief minutes that Sandra looked facts in the face before she took to demanding impossible things once more, she realized that she had backed into her own trap. She had been demanding. She had chosen to teach those who met her the Terran language instead of learning Telfan. Now those who understood any bit of Terran had gone to meet the *Haywire Queen*, leaving her among those who could not understand her at all. She could not communicate her desires to any of them.

She could not even tell them of the desire that they wanted to hear: That she wanted to leave.

The whole city would have broken a blood vessel to get her out.

But they didn't talk the same language.

The *Haywire Queen* came down in a screaming, wild landing. She rifled down out of the sky, careening. She slanted for a half mile, and then squared away and came plummeting down vertically. Inside, the accelerometer was making wild gyrations as Timkins fought the controls.



The whistling of the big ship's passage through the air slid down the audible scale as the velocity dropped. The ship slowed, and came to a perfect landing—

Twelve feet above the surface!

Like a slug of lead, the *Haywire Queen* poised for the barest instant, and then dropped the intervening distance. The landing plates sank into the soft soil of Telfu for several feet and the plates groaned, a rivet or two squeaked, and some welded joints disagreed. But spaceships are rigid structures, made for hard usage and considerable stresses and strains. It weathered the hard landing, though the angle was slightly cocked due to the unevenness of the turf's hardness. The *Haywire Queen* was still space-worthy.

"Rotten pilot," muttered Hammond.

"Terrible," agreed McBride.

"Look, you two grinning apes. I missed Telfu by exactly one hundred and forty-four inches. Twelve feet in 2,630,000,000,000,000 feet. Well within the experimental error, I think."

"Twelve feet in nine light-years isn't bad," said McBride. "Some day, Larry, you can bend that mathematical mechanism you use instead of a brain into calculating whether the landing effect would have been worse at *plus* twelve feet instead of minus."

"A mere matter of kinetic energy dissipated—"

"Yeah, we know. Well, you didn't kill us," laughed Hammond. "So let's go out and take a look at the wonders of the Telfan scenery."

"Take a quick look," said McBride. "Here come some Telfans to take a look at some Terran science."



"Wonder how they got here so quick," asked Timkins of no one in particular.

"Ask 'em."

Timkins stepped out of the space lock and smiled at the Telfans. "Ave, Canis Majoris," he said in a deep voice.

"Lousy Latin," snorted McBride.

"That's where they live."

"Do they know that?"

The foremost Telfan, who was Tet'h, stepped forward and smiled. "You ... Terrans?"

"Yes."

He pointed to the ship. "'Aywire Queen?"

"Yes."

Tet'h smiled once more and offered his hand.

"Universal gesture?" asked Hammond.

"No. Drake must have taught them that."

"Drake?" asked Tet'h. "You like?"

"Extremely doubtful," said Hammond. He was misunderstood. McBride said nothing but that pinching of the nose between thumb and forefinger conveyed the idea excellently.

"Telfans ... no like Drake."

"No?"

"No. Tall. Ugly-bald." Tet'h indicated his own luxurious pelt and then became confused as he realized that the Terrans were of the same, "Ugly-bald" complexion. He covered his face with both hands and muttered something that sounded apologetic and humble.

"Forget it," laughed McBride. "We ... like Telfans."

"Not like Drake," said Tet'h.

"Thanks," said Hammond honestly.

"How know ... here?" asked Timkins.

"You here?" asked Tet'h pointing to the ship and the surrounding landscape.

"Aren't we?" grinned Timkins.

"Save the fine rhetoric for later when they get the point of double talk," suggested Hammond.

Tet'h led them to the plane and Gormal and Elyon lifted a large case out. Tet'h opened it and handed McBride a little instrument. It was a cabinetless

job, every part exposed.

"Holy spinach," he said. "A mechanogravitic detector."

Hammond got a small mechanical planetarium showing Telfu and a minute sphere. Tet'h pulled a roller-map out of the base and indicated Telfu and the sphere. The map was a fairly accurate contour map of the blanketed region's contour.

Tet'h signified the cusp and then pointed to the position of Soaky. Below the cusp, Tet'h indicated the planet and then pointed to the ground.

"Here," he said.

McBride and Hammond tangled in an effort to shake Tet'h's hand. The Telfan looked proud.

"Many years," he said haltingly. "Work," indicating the detector. He made assembly motions. He pulled a book of mathematical identities from a pocket and said: "Found ... here." Then he made vast motions indicating a large construction. "Many years ... try like hell ... no work." He indicated the small satellite. "He make stop."

"Bright lads," grinned Hammond. "Their civilization was ready to discover the gravitic spectra. They did. They found it in math. They tried it and it didn't click too well. They discovered why. Never having anything of any great power operating, they never got to the point where they could build anything big enough to get off of Telfu. Just plain stuck. Well, fellers, if that moonlet is cupralum, I can see a lot of birds mining it."

"How're they going to land on it? Nothing gravitic will be worth a hoot that close."

"Lift 'em off the dead spot by battery-powered gravitics. Inefficient as hell. Get into space and then use rockets to land on that moonlet. Mine it. Load it full of detonite and blast."

"A hundred-mile moonlet?"

"They've got a nine-thousand-mile planet here to support it. They can't power their machinery with gravitrons, but electronics is an art worth remembering. One of the earlier atomic gadgets would do plenty."

"Might bore a large hole in it and pack in a mile of Atomite," suggested McBride. "I'd hate to support that, though."

"Better get some seetee meteors and pelt it by remote control," said Hammond. "Well, we can cover that later." To Tet'h he said: "You come in?"

Tet'h and Thani held a quick conference. "She come, too?" he asked.

"All of you."

"No. They stay. We go Terra."

"Terra!" exploded Hammond.

"Much to learn—both of us. You and I. You learn Telfan. We learn Terran. Better talk. This ... lousy."

"Easy to see Sandra's delicate hand in this language lesson," grinned Timkins.

"Better call that wild woman. Tell her we're going to take off in one hour and ten minutes because if we don't, we'll be as stuck as she is and we don't like that. As long as we have a bit of Telfu to take back with us in the shape of Tet'h and his woman Thani, we needn't stick around. I'll feel better about getting off on this rotation anyway. G'wan, we'll listen to you make the excuses, Larry."

"My turn to poke her on the pretty little schnozzola?"

"You won that by that three times something to the minus umpty-umpth power percentage of landing error. Twelve feet in what?"

"2,630,000,000,000,000,000 feet."

"Was that the same he said before?" asked McBride with a smile. "Or was he working that old gag about our not remembering?"

"I don't remember either."

"So, you win," said McBride to Larry Timkins.

Timkins called, and Sandra Drake's slightly hysterical voice replied.

"How you doing?" asked Larry.

"Where are you?"

"I don't know."

"Don't know?" said Sandra. Her voice went up in a crescendo and hit "G" above High "C" on the last note.

"No," said Larry. "Chicago, Venuland, Canalport, and Sharon are my best landmarks and they're all equally distant and in the same direction from here."

"Go to hell."

"That's across the River Styx from Sharon, on Pluto," said Timkins. "And that expression is making the Sharonites unhappy because people have been going there for thousands of years. Sharon hasn't the popularity."

"But look, Larry, I want to go along."

"Can you get here in one hour and eleven minutes. That's the absolute deadline until we can get to Terra and cook up a drive that's detuned enough from the cupralum-absorption region to permit us to tinker off and on around here."

"Where are you? How can I get there if you don't know where you are?"

"Ask someone."

Sandra's language became something that the communications commission has legislated against.

"Can you come here and get me?"

"We'll be doing fine if we get off with our skin," said Larry. "We definitely have not enough power to go roaming all over Telfu. We're on the one spot that will allow us to leave under the emergencies. An hour and thirty minutes from now that spot will be somewhere else. We'll wait an hour and ten and take off on the edge of the spot."

"Won't they come back and get me?"

"Wait a minute." Then he turned to Tet'h. "Could you send them back for Drake?"

"Yes," answered Tet'h. "Better not, though. She bad ... but lazy. Teach Terran so not ... learn Telfan."

"Sandra? No dice. That's it, toots. Take it or leave it."

"Look, Larry, isn't there something you can do?"

"I doubt it. Give you a tip, though. Next time you poke someone else's nose into a mess remember that he who laughs last isn't always too dumb to catch on quick. At the next sound, it will be exactly three people making with deep belly laughs. So long, until we meet again—in about six months! In-you, we're at these Telfan co-ordicidentally, if you should find someone who would like to get rid of nates: South Longitude.... Hey, Tet'h, how do you pronounce these figures?"

Tet'h caught his meaning and said: "Me tell."

He addressed the microphone, and spoke in Telfan. "There," he finished, "is where ... are!"

Timkins added: "So now you can get here all right."

He closed the mike as the speaker started to make little animal sounds. "Fellows," said Larry. "She's mad!"

"Crazy mad or angry mad?"

"Boiling mad."

"She'll be hard-boiled by the time she gets through stewing in her own juice," grinned Hammond. "Let's get some sky, fellows. O.K. ... we go?" he asked Tet'h.

"We go," said Tet'h cheerfully.

There was a quick conference between the two men who were to stay and Tet'h. Then the air-lock door was closed, and Timkins started to set up the controls.

Up in the emergency room, the batteries started to fume and fret as the terrible overload hit them. The *Haywire Queen* lifted uncertainly, gained a little speed, and then took off into the cloudless sky at an acceleration that varied continuously between nine to twenty feet per second per second per second under the super drive.

Not too long after, the gravitron-cupralum drive took over, and the *Haywire Queen* pointed her dome upwards at tiny Sol, blinking there in the sky between the constellations Aquila and Ophiuchus.

THE END.

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